Practical 8

A system has RAM which can store four pages simultaneously to satisfy page reguests. Write a Program to calculate percentage page hit and page fault if the system uses Optimal page replacement technique for the page references entered by the user i.e. 1, 2, 3, 4, 1, 2, 5, 1, 2, 3, 4, 5 (No of Frames=3).

Code:-

```
GNU nano 7.2
#include <stdio.h>

int main() {
    int pages[] = {1,2,3,4,1,2,5,1,2,3,4,5};
    int n = 12, frames = 3;
    int frame[3], time[3];
    int i, j, pos, hits = 0, faults = 0, counter = 0;

    // initialize frames
    for(i=0;i<frames;i++) {
        frame[i] = -1;
        time[i] = 0;
    }

    for(i=0;i<n;i++) {
        int found = 0;

        // check hit
        for(j=0;j<frames;j++) {
            if(frame[j] == pages[i]) {
                hits++;
                counter++;
                time[j] = counter;
                found = 1;
                break;
            }
        }

        // if page fault
        if(!found) {
            faults++;

            pos = -1;

            // check for empty frame first
            for(j=0;j<frames;j++) {
                if(frame[j] == -1) {
                    pos = j;
                    break;
                }
            }

            // if no empty frame → apply LRU
            if(pos == -1) {
                pos = 0;
                for(j=1;j<frames;j++) {
                    if(time[j] < time[pos])
                        pos = j;
                }
            }

            counter++;
            frame[pos] = pages[i];
            time[pos] = counter;
        }
    }

    printf("\nPage Hits = %d", hits);
    printf("\nPage Faults = %d", faults);
    printf("\nHit Percentage = %.2f%%", (float)hits/n*100);
    printf("\nFault Percentage = %.2f%%\n", (float)faults/n*100);

    return 0;
}
```

Output:

```
Page Hits = 2
Page Faults = 10
Hit Percentage = 16.67%
Fault Percentage = 83.33%
```